of a government, government -authorized hydrographic office or other relevant government institution, and conform to IHO standards as listed in appendix 1.

4.2 The contents of the system database should be adequate and up to date for the intended voyage to comply with SOLAS regulations V/19 and V/27.

4.3 It should not be possible to alter the contents of the ENDS or system database information transformed from the ENDS. The display of the content of ENDS should be compliant with IHO standards including rules set for interoperability.

4.4 ECDIS should be capable of accepting official updates to the ENDS provided in conformity with IHO standards. These updates should be automatically applied to the system database. By whatever means updates are received, the implementation procedure should not interfere with the display in use.

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database. By whatever means updates are received, the implementation procedure should not interfere with the display in use.

4.5 ECDIS should also be capable of accepting updates to the ENDS data entered manually with simple means for verification prior to the final acceptance of the data. They should be distinguishable on the display from ENDS i nformation and its official updates and not affect display legibility.

4.6 ECDIS should keep and display on demand a record of updates including time of application to the system database. This record should include updates for each ENDS until it is superseded by a new edition.

4.7 ECDIS should allow the mariner to display updates in order to review their contents and to ascertain that they have been included in the system database.

4.8 ECDIS should be capable of accepting ENDS in accordance with the I HO Data Protection Scheme.3

MODULE B OPERATIONAL AND FUNCTIONAL REQUIREMENTS

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Protection Scheme.3

MODULE B OPERATIONAL AND FUNCTIONAL REQUIREMENTS

5 DISPLAY OF SYSTEM DATABASE INFORMATION

5.1 An ECDIS should be capable of accepting and converting an ENDS and their updates into a system database. ECDIS should be capable of displaying and processing all system database information as specified by IHO. The ECDIS may also be capable o f accepting a system database resulting from conversion ashore, in accordance with IHO resolutions.4

5.2 System database information available for display during route planning and route monitoring should be subdivided into the following three categories: display base, standard display and all other information (see appendix 2).

5.3 ECDIS should present the standard display at any time by a single operator action.

3 IHO Publication S -63 Data Protection Scheme (for S -57 ENCs) and S -100, Part 15 Data Protection Scheme (for S-100 products) (see appendix 1).

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Scheme (for S-100 products) (see appendix 1).

4 IHO Publication M-3 Resolutions of the IHO.
RESOLUTION MSC.530(106) (adopted on 7 November 2022)
PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS)
MSC 106/19/Add.1
Annex 22, page 6

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5.4 When an ECDIS is switched on following a switch off or power failure, it should return to the most recent manually selected settings for display.

5.5 It should be easy to add or remove information from the ECDIS display. It should not be possible to remove information contained in the display base.

5.6 For any operator -identified geographical position (e.g. by cursor picking), ECDIS should display on demand the information about the chart obje cts associated with such a position.

5.7 It should be possible to change the display scale by appropriate steps, e.g. by means of either chart scale values or ranges in nautical miles.

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of either chart scale values or ranges in nautical miles.

5.8 It should be possible for the mariner to select a safety contour from the information provided by the system database. ECDIS should emphasize the safety contour over other contours on the display. However:

.1 if the mariner does not specify a safety contour, this should default to 30 m. If the safety contour specified by the mariner or the default 30 m contour is not in the displayed system database, the safety contour shown should default to the next deeper contour;

.2 if the safety contour in use becomes unavailable due to a change in source data, the safety contour should default to the next deeper contour;

.3 in each of the above cases, an indication should be provided; and

.4 the mariner should be able to select a permanent display of safety contour and safety depth settings.

5.9 It should be possible for the mariner to select a safety depth. ECDIS should

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and safety depth settings.

5.9 It should be possible for the mariner to select a safety depth. ECDIS should
emphasize soundings equal to or less than the safety depth whenever spot soundings are selected for display.

5.10 It should be possible to use dynamic water level adjustment and an indication should be provided.

5.11 The ENDS and all updates to it should be displayed without any degradation of their information content.

5.12 ECDIS should provide a means to ensure that the ENDS and all updates to it h ave been correctly loaded into the system database.

5.13 The ENDS data and updates to it should be clearly distinguishable from other displayed information, including those listed in appendix 3.

6 SCALE

6.1 ECDIS should provide an indication if:

.1 the information is displayed at a larger scale than that contained in the ENC;

.2 own ship's position is covered by an ENC at a larger scale than that provided by the display; or

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.2 own ship's position is covered by an ENC at a larger scale than that provided by the display; or

.3 information at own ship's position is not displayed because of applying scale minimum for display.

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7 DISPLAY OF OTHER NAVIGATIONAL INFORMATION

7.1 Radar information and/or AIS information may be transferred from systems compliant with the relevant standards of the Organization. Other navigational information may be added to the ECDIS display. However, it should not degrade the displayed system database information and it should be clearly distinguishable from the system database information.

7.2 It should be possible to remove the radar information, AIS information and other navigational information by single operator action.

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7.2 It should be possible to remove the radar information, AIS information and other navigational information by single operator action.

7.3 ECDIS and added navigational information should use a common reference system. If this is not the case, an indication should be provided.

7.4 Radar

7.4.1 Transferred radar information may contain a radar image and/or tracked target information.

7.4.2 If the radar image is added to the ECDIS display, the chart and the radar image should match in scale, projection and orientation.

7.4.3 The radar image and the position from the position sensor should both be adjusted automatically for antenna offset from the conning position.

8 DISPLAY MODE AND GENERATION OF THE NEIGHBOURING AREA

8.1 It should always be possible to display the system database information in a "north-up" orientation. Other orientations are permitted. When such orientations are displayed,
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